

Element F: Consideration of Design Viability

There are several extra considerations that we must account for in our design, as listed below:

- 1) Part Costs - Our design will need at least two relatively powerful motors, a breadboard, some sort of LiPo battery, and a microcontroller. Fortunately, members of the team already have all of the above components, except for the motors. The motors themselves can be either sourced from pre-existing products (electric wheelchairs, e-bikes, etc) or purchased with some sort of additional funding.
- 2) Time - Our design will take significantly longer to design, acquire parts, and build due to the added challenge of the electrical component and 3D printing the chassis. We have about 10 school days left to complete the project, so, with 3 team members and 47 minutes per class period, factoring in a 60% efficiency rate in class, we have about 13.5 hours of manpower to complete and document our project. While we can, and likely will, work on the project outside of class, all team members have work from other classes and extracurricular commitments (debate, robotics, cross country, etc), so we should aim to complete most of the work inside class time. This may be the largest issue with our current design.
- 3) Ergonomics - Our design is probably one of the most ergonomic designs we have seen so far, as it can eliminate the need for the user to push the cart. However, the user will still have to bend over to attach the pod to the cart, meaning the user still could have issues using the product. We can mitigate this by making the latches as easy as possible to connect, or even have some sort of attachment mechanism that can be operated with a foot.